Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

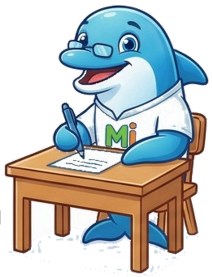
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Code the pattern	Not in this proof edition				☐
2	Prime hunters	Not in this proof edition				☐
3	Below zero	Not in this proof edition				☐
4	Algebra machines	Not in this proof edition				☐
5	Every possible way	Not in this proof edition				☐
6	Fair or unfair?	6-7	8-9	10 and 12	11	☐
7	Turtle geometry	Not in this proof edition				☐
8	Secret codes	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Showcase challenge	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

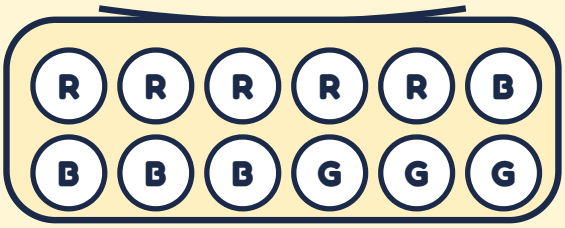
Learn it: Probability

My goal. I can write a probability as a fraction, decide whether a game is fair, and say what to expect from many tries.

A probability is the number of ways you want, out of all the equally likely outcomes. A game is fair when every player has the same probability of winning. In many tries we expect about probability × tries, but a real run varies.

Worked example

A bag has 12 counters: 5 red (R), 4 blue (B) and 3 green (G). One is picked without looking. What is the probability of blue? About how many blues in 600 picks, if the counter goes back each time?



1. All 12 counters are equally likely to be picked.
2. 4 of the 12 are blue, so P(blue) = 4/12 = 1/3.
3. In 600 picks we expect 1/3 × 600 = 200 blues.
4. A real run gives a number near 200, not exactly 200. Each pick is random.

Answer: P(blue) = 1/3. About 200 blues in 600 picks.

How to find a probability

1. Count all the equally likely outcomes.
2. Count the outcomes you want.
3. Write wanted/all as a fraction and simplify it.
4. Expected number = probability × number of tries. A real run is close, not exact.

Your turn

Answer at the back of the book.

Try **A bag has 6 yellow and 9 blue counters. One is picked without looking.**

P(yellow) = ☐ / ☐ in simplest form.

Start here

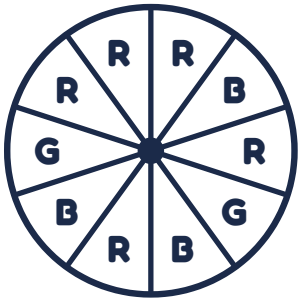
Do these first. Take your time.

1 **The spinner has 10 equal parts. R is red, B is blue and G is green. Give fractions in simplest form.**

(a) P(green) = ☐ / ☐

(b) P(not red) = ☐ / ☐

(c) In 500 spins, about ☐ are blue.



2 **A coin is flipped and a spinner with 3 equal parts A, B and C is spun. A result looks like HB. Use the table to list the results. Give fractions in simplest form.**

	A	B	C
H			
T			

(a) There are ☐ different results. (b) P(heads and A) = 1/☐.

(c) P(tails and not A) = ☐ / ☐

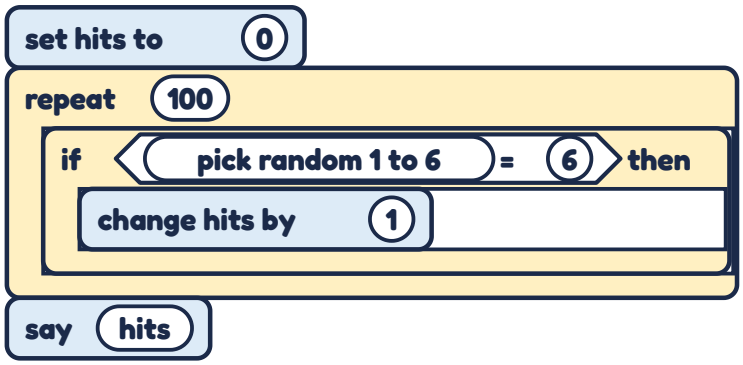
3 **This code rolls a dice 100 times and counts the sixes. Predict what happens.**

(a) The chance of a 6 on one roll is ☐ / ☐.

(b) The code says about ☐.

(c) 5 runs give 14, 19, 17, 21, 15. Why?

random rolls broken code



Start here

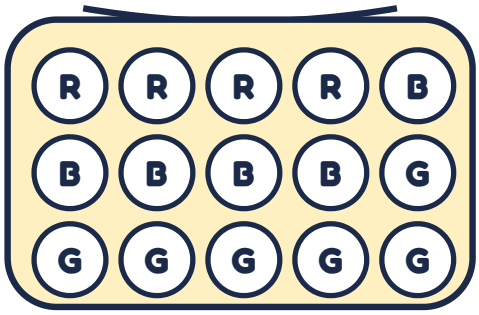
Keep going. Check your work.

4 **One counter is picked from the bag. R is red, B is blue and G is green. Give fractions in simplest form.**

(a) P(blue) = ☐ / ☐

(b) P(red or green) = ☐ / ☐

(c) 5 more red counters are put in. Now P(red) = ☐ / ☐ .

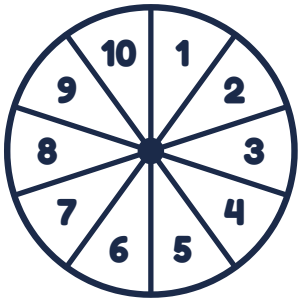


5 **Ali and Ben spin this spinner with the numbers 1 to 10. Ali wins on a prime number. Ben wins on any other number.**

(a) P(Ali wins) = ☐ /10

(b) P(Ben wins) = ☐ /10

(c) Is the game fair? Yes No



6 **A dice is rolled 300 times and a coin is flipped 90 times. Give the number you expect.**

(a) About ☐ fives. (b) About ☐ numbers greater than 4.

(c) About ☐ heads in 90 flips.

(d) A run gives 54 fives. Is that surprising? Yes No

7 **Bag A has 3 red and 5 blue counters. Bag B has 4 red and 7 blue counters. You want to pick a red counter.**

(a) In Bag A, P(red) = ☐ / ☐ .

(b) In Bag B, P(red) = ☐ / ☐ .

(c) Which bag gives the better chance of red? Bag A Bag B

Start here

Nearly there. You can do it.

8 **Two dice, red and blue, are rolled and added. The table shows all 36 equally likely totals. Give fractions in simplest form.**

(a) ☐ of the 36 totals are 4.

(b) ☐ of the 36 totals are less than 5.

(c) P(total is 4) = ☐ / ☐

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

9 **Cards are numbered 1 to 30. One card is drawn without looking. Give fractions in simplest form.**

(a) P(multiple of 7) = ☐ / ☐

(b) P(prime) = ☐ / ☐

(c) P(square number) = ☐ / ☐

10 **Ravi flips a fair coin 200 times and gets 112 heads.**

(a) About ☐ heads are expected.

(b) The difference from the expected number is ☐.

(c) Could the coin still be fair? Yes No

11 **Two dice are rolled and added. The totals 2 to 12 are not equally likely.**

(a) The most likely total is ☐.

(b) P(total is 6 or 8) = ☐/36

(c) P(total is 2 or 12) = ☐/36

Think harder

Hints are upside down. Try first!

1 **Two dice are rolled and added. In Game 1, Ali wins on an even total and Ben wins on an odd total.**

(a) Game 1: Ali wins on ☐ of the 36 totals.

(b) Is Game 1 fair? Yes No

(c) Game 2: Ali wins on a total of 7. Ben wins on every other total and scores 1 point for a win. For a fair game Ali must score ☐ points for a win.

Working: _____

Hint: Fair means the winning points are equal over many games.

2 **A bag has 3 red counters (R1, R2, R3) and 2 blue counters (B1, B2). Two counters are taken out one after another. The first is not put back. Give fractions in simplest form.**

(a) There are ☐ equally likely ordered pairs.

(b) ☐ of them are both red. (c) P(both red) = ☐ / ☐

Working: _____

Hint: List the pairs: R1R2, R1R3, R1B1, … Is R2R1 different?

3 **A program rolls a dice 600 times and counts the 3s. Five runs give 93, 108, 97, 114 and 102.**

(a) The expected number of 3s is ☐.

(b) The mean of the 5 runs is ☐.

(c) Does the dice look fair? Yes No

Working: _____

Hint: Add the 5 numbers, then divide by 5.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 6: Probability of at least one.

A coin is flipped 4 times.

(a) There are ☐ equally likely outcomes.

(b) P(exactly 2 heads) = ☐/16 (c) P(at least one tail) = ☐/16

Working: _____

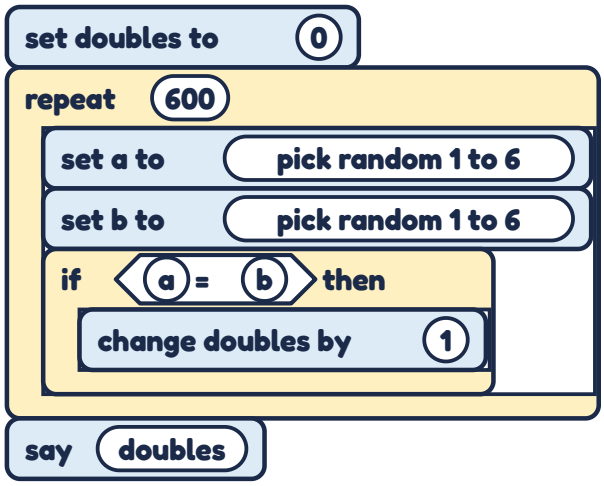
Hint. Which single outcome has no tails? Take it from all 16.

2 This code rolls 2 dice 600 times. Predict what it says.

(a) For a given a, b matches it in ☐ out of 6 ways.

(b) It says about ☐.

(c) Change a = b to a + b = 4. It says about ☐.



Hint. List the pairs (a, b) that add to 4.

3 Beyond Level 6: Expected winnings.

A game costs 2 dollars to play. You roll a dice and win a prize if you roll a 6. You play 600 times.

(a) You win about ☐ times.

(b) You pay ☐ dollars. With a 5-dollar prize you win about ☐ dollars.

(c) The game is fair when the prize for a 6 is ☐ dollars.

Working: _____

Hint. Fair means the money won equals the money paid.

✔ Check yourself

No hints. Try each one on your own.

1 A bag has 7 red and 3 blue counters. P(blue) = ☐ / ☐.

Working: _____

2 A dice is rolled 240 times. About ☐ of the rolls are even numbers.

Working: _____

3 Two coins are flipped. P(2 tails) = ☐ / ☐.

Working: _____

4 A spinner has 8 equal parts and 2 are green. In 400 spins about ☐ are green.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 6.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 6 · Fair or unfair?

Your turn
page 6

Answer: 2/5
Why: 15 counters in all and 6 are yellow: 6/15. Divide the top and bottom by 3: 2/5.

START HERE

1
page 7

Answer: (a) 1/5 (b) 1/2 (c) 150
Why: There are 10 equally likely parts. Green: 2 parts, so 2/10 = 1/5. Not red: 3 blue + 2 green = 5 parts, so 5/10 = 1/2. Blue is 3/10, and 3/10 × 500 = 150.

2
page 7

Answer: (a) 6 (b) 6 (c) 1/3
Why: The coin has 2 results and the spinner has 3, so 2 × 3 = 6 equally likely results, one for each cell. HA is 1 of the 6, so 1/6. Tails with B or C: TB and TC, so 2/6 = 1/3.

3
page 7

Answer: (a) 1/6 (b) 17 (c) random rolls
Why: One roll is 6 with probability 1/6. In 100 rolls we expect 1/6 × 100 = 16.7, so about 17. The answers differ because every roll is random, so each run gives a slightly different count.

4
page 8

Answer: (a) 1/3 (b) 2/3 (c) 9/20
Why: The bag has 4 + 5 + 6 = 15 counters. Blue: 5/15 = 1/3. Red or green: 4 + 6 = 10, so 10/15 = 2/3. With 5 more red counters there are 9 red out of 20: 9/20.

5
page 8

Answer: (a) 4 (b) 6 (c) No
Why: The primes from 1 to 10 are 2, 3, 5 and 7, so Ali wins on 4 of the 10 numbers. Ben wins on the other 6 (1, 4, 6, 8, 9, 10). 4 is not equal to 6, so the game is not fair: Ben is more likely to win.

6
page 8

Answer: (a) 50 (b) 100 (c) 45 (d) No
Why: (a) 1/6 × 300 = 50. (b) 5 or 6 is 2 out of 6, so 1/3 × 300 = 100. (c) 1/2 × 90 = 45. (d) 54 is close to 50, so it is not surprising. A random run is rarely exactly the expected number.

7
page 8

Answer: (a) 3/8 (b) 4/11 (c) Bag A
Why: Bag A: 3 red out of 8, so 3/8. Bag B: 4 red out of 11, so 4/11. Compare with a common bottom number, 88: 3/8 = 33/88 and 4/11 = 32/88. 33 is greater than 32, so Bag A gives the better chance.

8
page 9

Answer: (a) 3 (b) 6 (c) 1/12
Why: Count in the table. Total 4 is 1+3, 2+2 and 3+1: 3 totals. Totals less than 5 are 2, 3 and 4: 1 + 2 + 3 = 6 totals. 3 out of 36 is 1/12.

9
page 9

Answer: (a) 2/15 (b) 1/3 (c) 1/6
Why: There are 30 equally likely numbers. Multiples of 7: 7, 14, 21, 28 is 4, so 4/30 = 2/15. Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 is 10, so 10/30 = 1/3. Squares: 1, 4, 9, 16, 25 is 5, so 5/30 = 1/6.

10 page 9
Answer: (a) 100 (b) 12 (c) Yes
Why: (a) 1/2 × 200 = 100. (b) 112 − 100 = 12. (c) A random run is not exactly 100 each time. 112 is not far from 100, so the coin could still be fair.

11 page 9
Answer: (a) 7 (b) 10 (c) 2
Why: The totals 2 to 12 are not equally likely. 7 happens 6 ways, which is the most. 6 happens 5 ways and 8 happens 5 ways: 10/36. 2 (1+1) and 12 (6+6) happen 1 way each: 2/36.

THINK HARDER

1 page 10
Answer: (a) 18 (b) Yes (c) 5
Why: An even total needs both dice odd (3 × 3 = 9 ways) or both dice even (3 × 3 = 9 ways): 18 of the 36. So 18/36 each: fair. Game 2: Ali wins on 7, which is 6/36, and Ben on the other 30/36. If Ali scores 5 points a win and Ben scores 1, then 6 × 5 = 30 and 30 × 1 = 30: fair.

2 page 10
Answer: (a) 20 (b) 6 (c) 3/10
Why: The first counter has 5 choices and the second has 4: 5 × 4 = 20 equally likely ordered pairs. Both red: 3 choices, then 2: 6 pairs. 6/20 = 3/10.

3 page 10
Answer: (a) 100 (b) 102.8 (c) Yes
Why: (a) 1/6 × 600 = 100. (b) 93 + 108 + 97 + 114 + 102 = 514, and 514 ÷ 5 = 102.8. (c) Every run is near 100 and the mean is very close to 100, so the dice looks fair.

GO FURTHER

1 page 11
Answer: (a) 16 (b) 6 (c) 15
Why: Each flip has 2 results, so 2 × 2 × 2 × 2 = 16 equally likely outcomes. Exactly 2 heads: HHTT, HTHT, HTTH, THHT, THTH, TTHH is 6, so 6/16. At least one tail is every outcome except HHHH: 15/16.

2 page 11
Answer: (a) 1 (b) 100 (c) 50
Why: For any a, only 1 of the 6 values of b matches it: 1 out of 6. So we expect 1/6 × 600 = 100 doubles. The pairs that add to 4 are 1+3, 2+2 and 3+1: 3 of 36, which is 1/12. 1/12 × 600 = 50.

3 page 11
Answer: (a) 100 (b) 1200, 500 (c) 12
Why: A 6 comes up about 1/6 × 600 = 100 times. You pay 600 × 2 = 1200 dollars. If the prize is 5 dollars you win about 100 × 5 = 500 dollars, so you lose money. The game is fair when the winnings equal what you paid: 100 × prize = 1200, so the prize is 12 dollars.

CHECK YOURSELF

1 page 12
Answer: 3/10
Why: 10 counters in all and 3 are blue: 3/10. It cannot be simplified.

2 page 12
Answer: 120
Why: Even numbers are 2, 4 and 6, so the probability is 3/6 = 1/2. 1/2 × 240 = 120.

3
page 12

Answer: 1/4
Why: The outcomes are HH, HT, TH and TT, which are equally likely. Only TT has 2 tails: 1/4.

4
page 12

Answer: 100
Why: P(green) = 2/8 = 1/4. 1/4 × 400 = 100.